Capture the Flag 5: Make the System Interactive

Goal: Transform your Student Directory into an interactive application where administrators can favorite students, edit information, and remove records using Flutter interactions and state management.

Instructor Note

These activities require your own problem-solving and research skills.

DO NOT use AI to generate the code for you.

You may use AI to:

- Research what a Dart/Flutter feature does
- Understand syntax
- Explain an error (not fix your error)
- Clarify documentation

You may NOT ask AI to generate the solution/code for a flag.

🚩 Flag 0 — Create Your Feature Branch

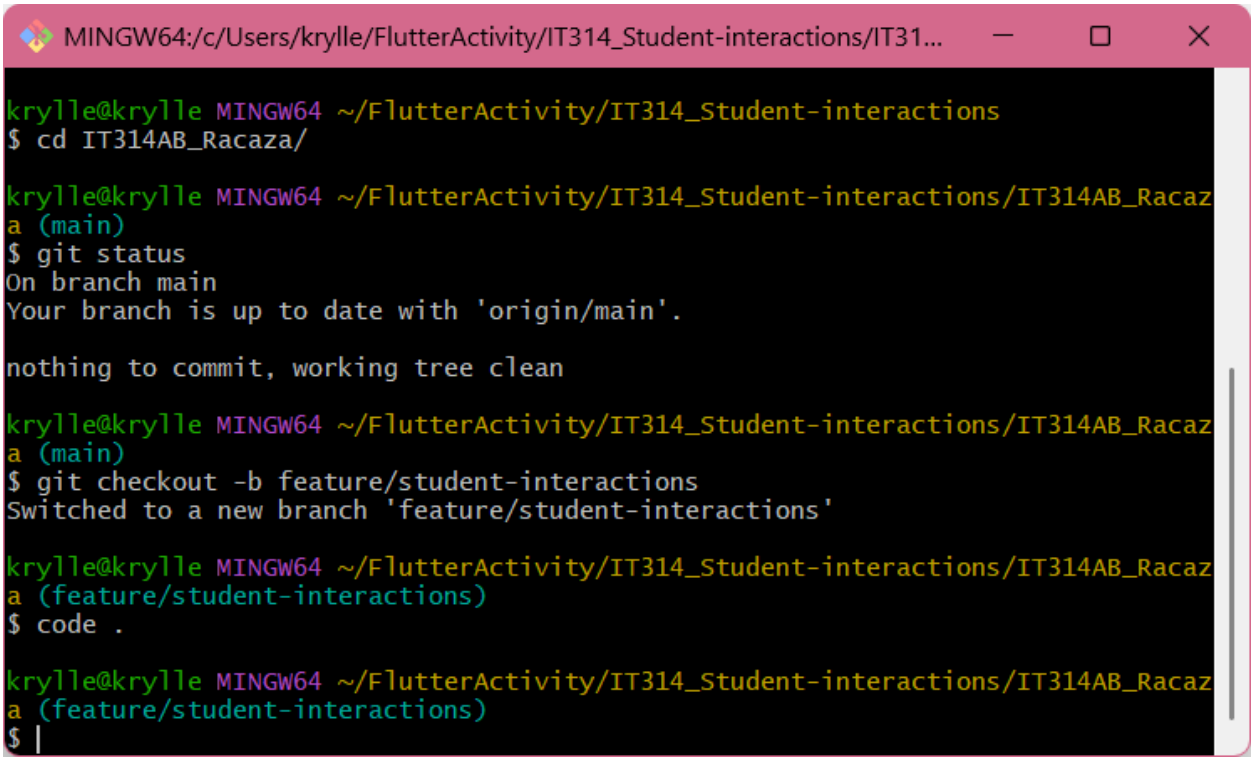
Objective: Create a new feature branch before adding interactions.

Create:

`feature/student-interactions`

Screenshot:

- Terminal showing your current branch



🚩 Flag 1 — Wake Up the Buttons

Add at least two buttons inside each Student Card.

Example actions:

- Favorite
- Edit

Objective: Make your Student Card respond to button presses using `onPressed`.

Right now your buttons are only decorations.

Research how Flutter's `onPressed` works.

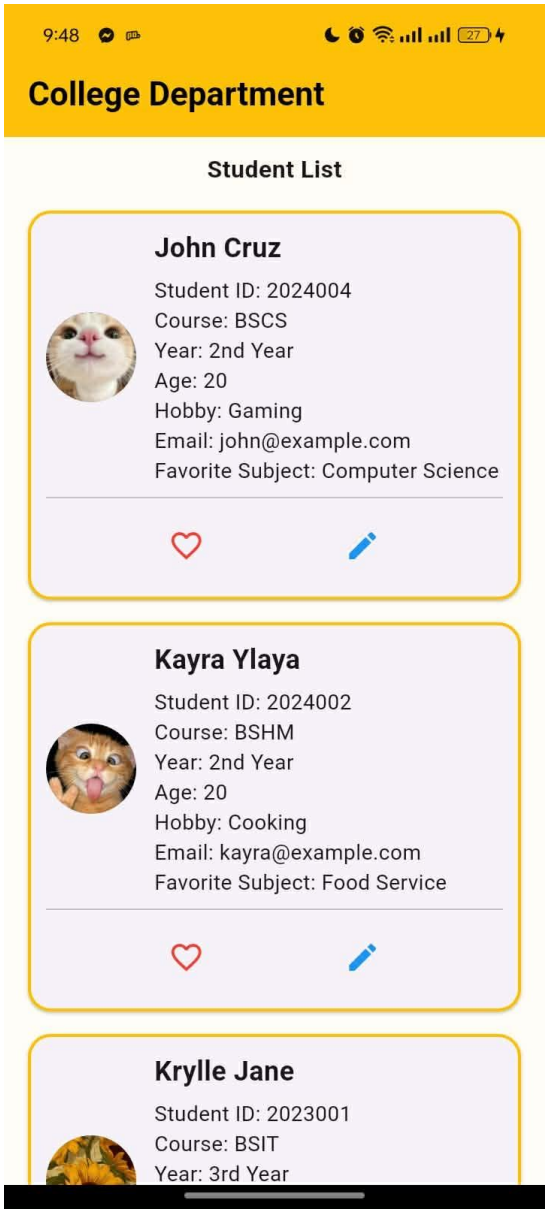
The buttons do not need to perform their final behavior yet. They simply need to respond when pressed.

Test

When a button is pressed, your app should visibly react in some way (such as printing a message or showing a temporary notification).

Screenshot:

- Student Card with buttons
- `onPressed` inside your code
- Button reacting or terminal message




```dart
const Divider(),

Row(
  mainAxisAlignment:
      MainAxisAlignment.spaceEvenly,

  children: [
    IconButton(
      icon: const Icon(
        Icons.favorite_border,
        color: Colors.red,
      ), // Icon

      onPressed: () {
        print('student $student Favorite pressed');
      },
    ), // IconButton
    IconButton(
      icon: const Icon(
        Icons.edit,
        color: Colors.blue,
      ), // Icon

      onPressed: () {
        print('Student $student edit pressed');
      },
    ), // IconButton
```

```
Problems 3  Output  Debug Console  Terminal

Terminal

PS C:\Users\krylle\FlutterActivity\IT314_Student-interactions\IT314AB_Racaza\my_first_flutter_app> flutter run
D/ViewRootImplExtImpl( 9083): the up motion event handled by client, just return
I/flutter ( 9083): student {image: assets/profile2.png, name: Kayra Ylaya, course: BSHM, year: 2nd Year, age: 20, hobby: Cooking, studentId: 2024002, email: kayra@example.com, favoriteSubject: Food Service} Fa
ite pressed
D/ViewRootImplExtImpl( 9083): the up motion event handled by client, just return
I/Quality ( 9083): Skipped: false 1 cost 26.113707 refreshRate 16849596 bit true processName com.example.my_first_flutter_app
D/ViewRootImplExtImpl( 9083): the up motion event handled by client, just return
I/flutter ( 9083): Student {image: assets/profile1.png, name: Krylle Jane, course: BSIT, year: 3rd Year, age: 20, hobby: Dancing, studentId: 2023001, email: krylle@example.com, favoriteSubject: Programming} ed
pressed
```

🚩 Flag 2 — Touch Anywhere

Objective: Make the entire Student Card tappable using `onTap`.

Buttons aren't the only interactive elements.

Research Flutter's `onTap`.

Wrap your Student Card so tapping anywhere on the card triggers an interaction.

Test

The user should be able to:

- Tap the Favorite button
- Tap the Edit button
- Tap anywhere else on the Student Card

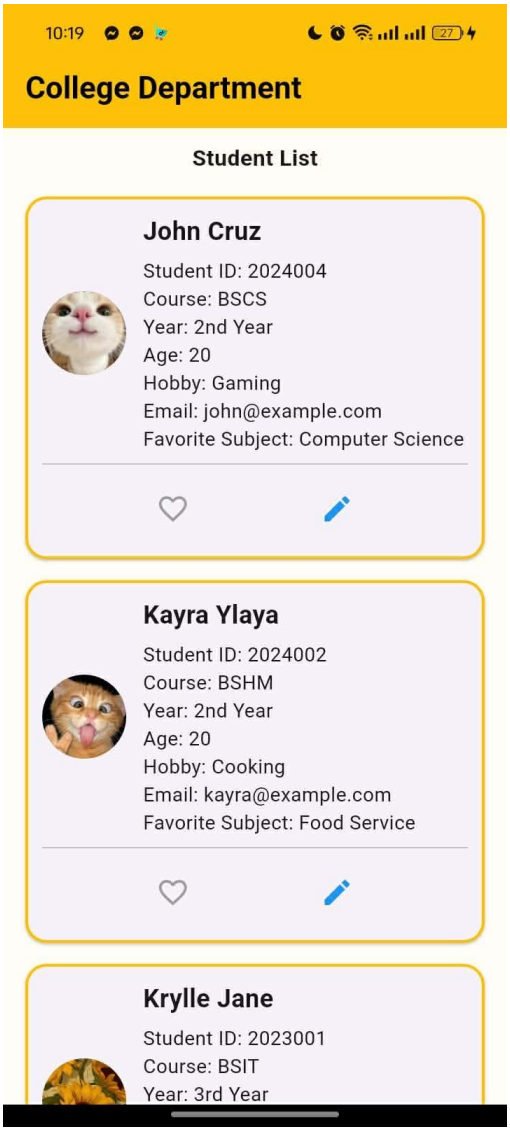
Each interaction should be recognized separately. (One function for the 2 buttons, another for the Student Card)

Screenshot:

- `onTap` in your code
- Student Card being tapped

```dart
return GestureDetector(
  onTap: () {
    print ('student card tapped');
  },
```

```
PS C:\Users\krylle\FlutterActivity\IT314_Student-interactions\IT314AB_Racaza\my_first_flutter_app> flutter run
D/ViewRootImplExtImpl( 4382): the up motion event handled by client, just return
I/flutter ( 4382): student card tapped
D/ViewRootImplExtImpl( 4382): the up motion event handled by client, just return
I/flutter ( 4382): student card tapped
D/ViewRootImplExtImpl( 4382): the up motion event handled by client, just return
I/flutter ( 4382): student card tapped
D/ViewRootImplExtImpl( 4382): the up motion event handled by client, just return
I/flutter ( 4382): student card tapped
```



🚩 Flag 3 — Discover `setState`

Objective: Update the screen immediately using `setState`.

So far your app reacts. But does the UI actually change?

Research Flutter's `setState`.

Use it so pressing a button immediately updates something visible on the screen.

Examples include:

- Changing text
- Changing an icon
- Updating a counter

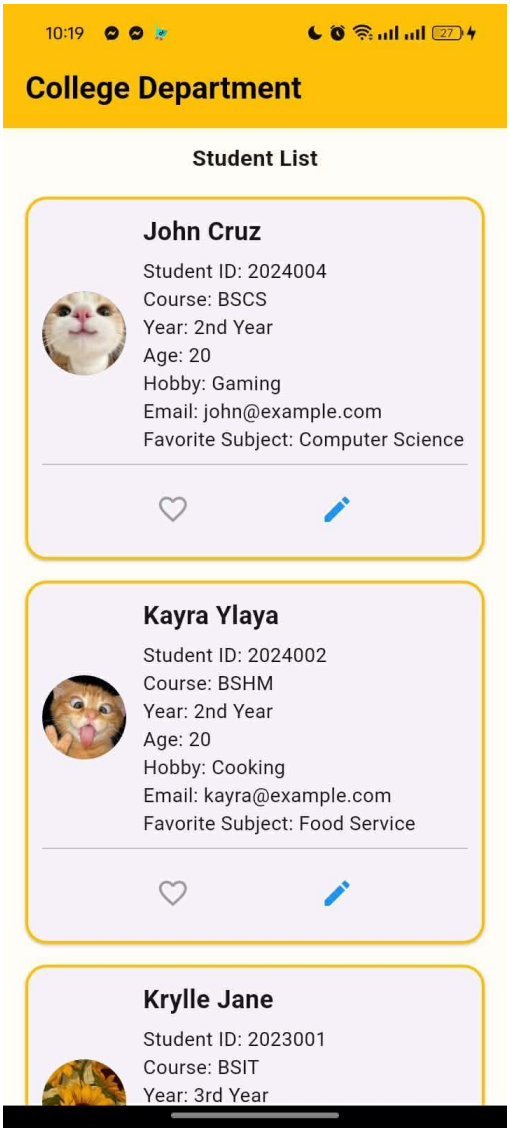
The important part is understanding that `setState` tells Flutter to rebuild the UI.

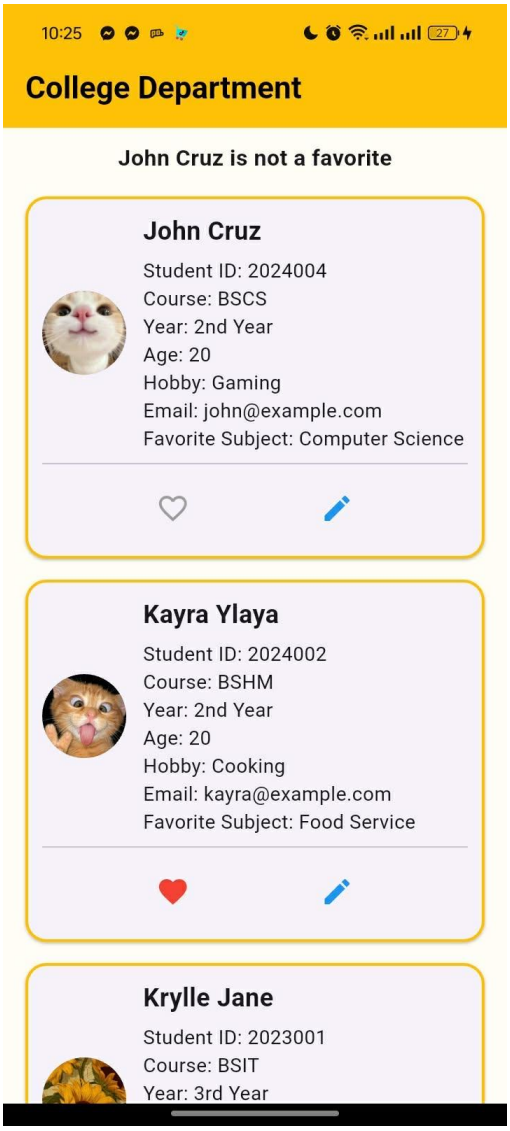
Test

The screen should visibly change immediately after pressing the button.

Screenshot:

- `setState` in your code
- Before and after interaction






```
  onPressed: () {
    setState(() {
      student['favorite'] =
          !student['favorite'];

      message =
          student['favorite']
              ? '${student['name']} is a favorite':'${student['name']} is not a favorite';
    });
  },
), // IconButton
```

🚩 Flag 4 — Favorite Students

Objective: Let administrators mark students as favorites.

Schools often need to highlight important records.

Add a favorite feature that changes a student's appearance when marked.

Possible visual changes include:

- Filled heart/star
- Different card color
- Favorite label

Each student's favorite status should work independently.

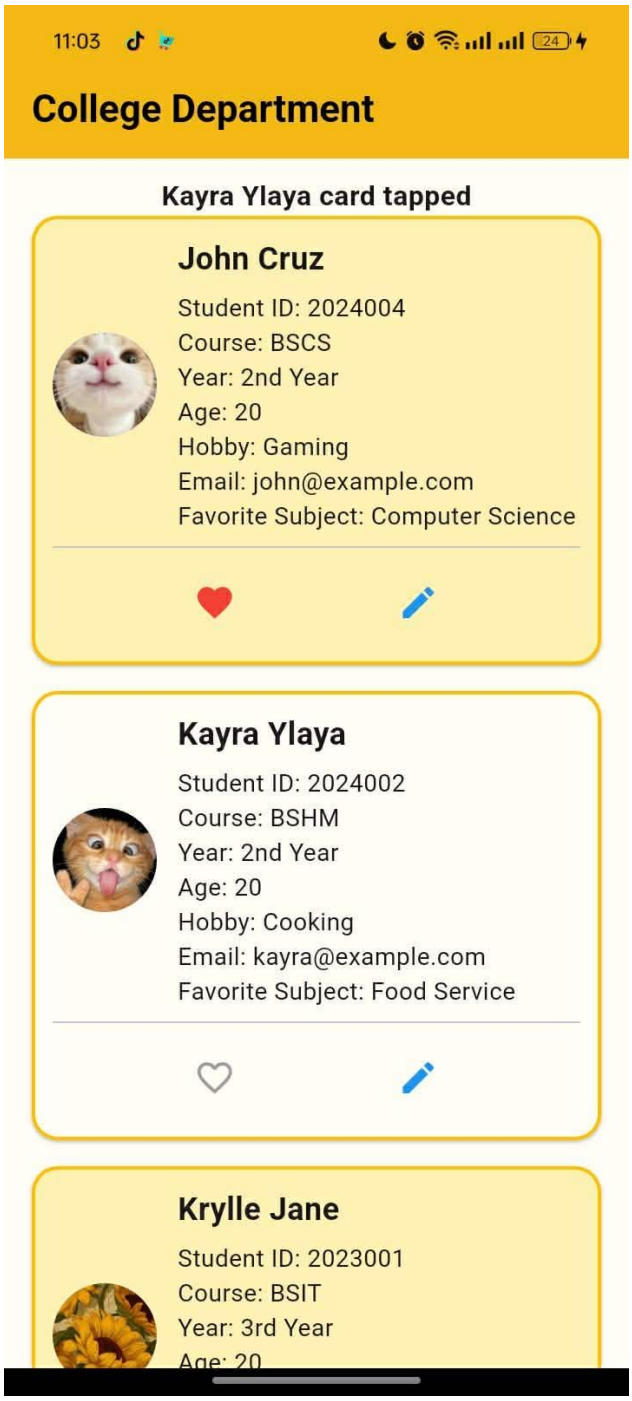
Test

- Favorite one student.
- Other students should remain unchanged.
- Favorite another student.

Both should keep their own state.

Screenshot:

- Favorited student
- Non-favorited student
- Favorite icon changing




```
child: Card(
  margin: const EdgeInsets.only(
    bottom: 15,
  ), // EdgeInsets.only

  color: student['favorite']
      ? const Color(0xFFFFF0B3)
      : const Color(0xFFFFFDF5),

  shape: RoundedRectangleBorder(
    borderRadius:
        BorderRadius.circular(15),

    side: const BorderSide(
      color: Color(0xFFFFC107),
      width: 2,
    ), // BorderSide
  ), // RoundedRectangleBorder
```

🚩 Flag 5 — Remove a Student

Objective: Delete a student from the directory.

Research how to remove an item from a Dart `List`.

Add a Delete action that removes a student from your list.

The UI should immediately update after deletion.

Test

Before:

- Student A
- Student B
- Student C

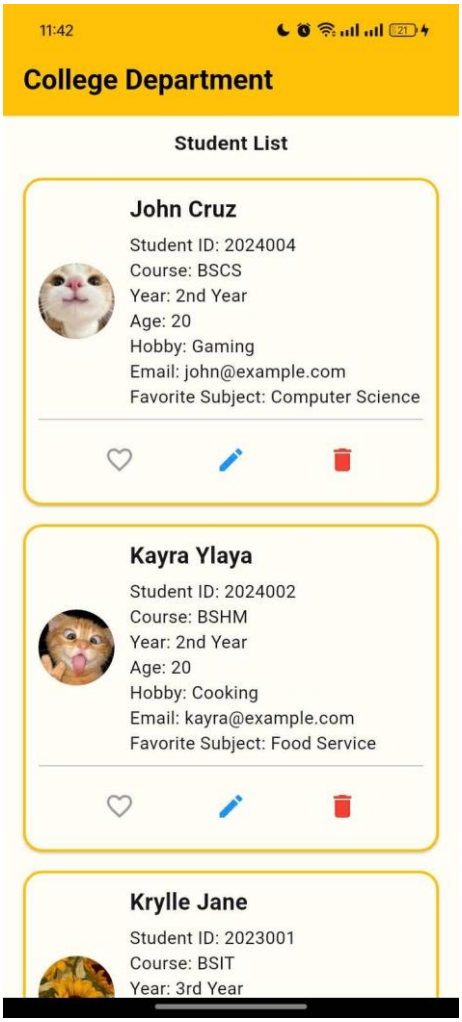
After deleting Student B:

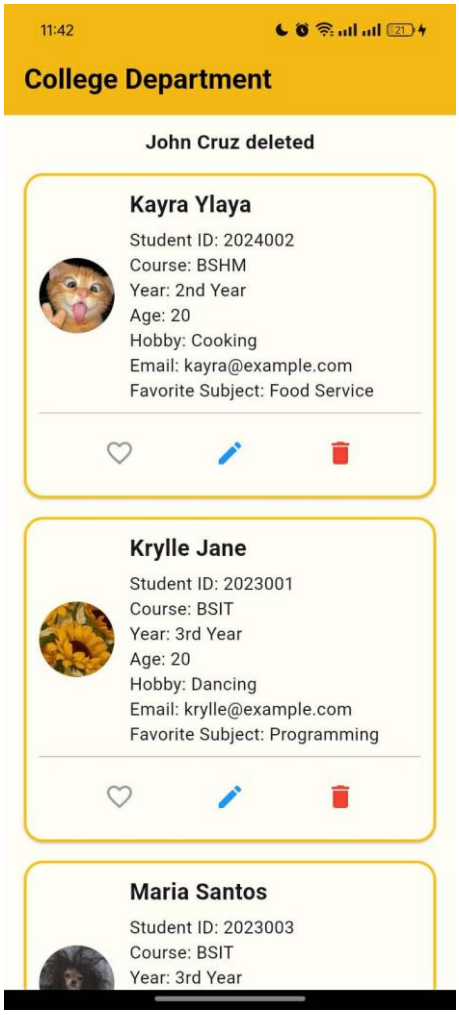
- Student A
- Student C

No empty spaces should remain.

Screenshot:

- Student before deletion
- Student after deletion






```
IconButton(
  icon: const Icon(
    Icons.delete,
    color: Colors.red,
  ), // Icon

  onPressed: () {
    setState(() {
      students.remove(student);

      message =
          '${student['name']} deleted';
    });
  },
), // IconButton
```

🚩 Flag 6 — Edit Trigger

Objective: Prepare your app for editing student information.

In the next CTF you'll build a full Edit Screen.

For now, create an Edit trigger.

Research simple ways Flutter can display temporary editing interfaces.

Examples include:

- `AlertDialog`
- `SnackBar`
- Bottom Sheet

The goal is simply to launch an edit action, not build the complete editor yet.

Test

Pressing Edit should open your chosen interface.

Screenshot:

- Edit button
- Opened dialog, snackbar, or bottom sheet

11:51

College Department

Student List

John Cruz



Student ID: 2024004
Course: BSCS
Year: 2nd Year
Age: 20
Hobby: Gaming
Email: john@example.com
Favorite Subject: Computer Science

Kayra Ylaya



Student ID: 2024002
Course: BSHM
Year: 2nd Year
Age: 20
Hobby: Cooking
Email: kayra@example.com
Favorite Subject: Food Service

Krylle Jane



Student ID: 2023001
Course: BSIT
Year: 3rd Year

11:51

College Department

Student List

John Cruz

Student ID: 2024004
Course: BSCS
Year: 2nd Year
Age: 20
Hobby: Gaming
Email: john@example.com
Favorite Subject: Computer Science

Edit Student

You selected John Cruz for editing.

Close

Age: 20
Hobby: Cooking
Email: kayra@example.com
Favorite Subject: Food Service

Krylle Jane

Student ID: 2023001
Course: BSIT
Year: 3rd Year


IconButton(
icon: const Icon(
Icons.edit,
color: Colors.blue,
), // Icon
onPressed: () {
showDialog(
context: context,
builder: (context) {
return AlertDialog(
title: const Text(
'Edit Student',
), // Text
content: Text(
'You selected ${student['name']} for editing.',
), // Text
actions: [
TextButton(
onPressed: () {
Navigator.pop(
context,
);
},
child: const Text(
'Close',
), // Text

🚩 Flag 7 — Multiple Actions, Independent State

Objective: Make every Student Card manage interactions correctly.

This is your final interaction challenge.

Verify that every student behaves independently.

Example scenario:

- Favorite Student A.
- Delete Student C.
- Open Edit for Student B.

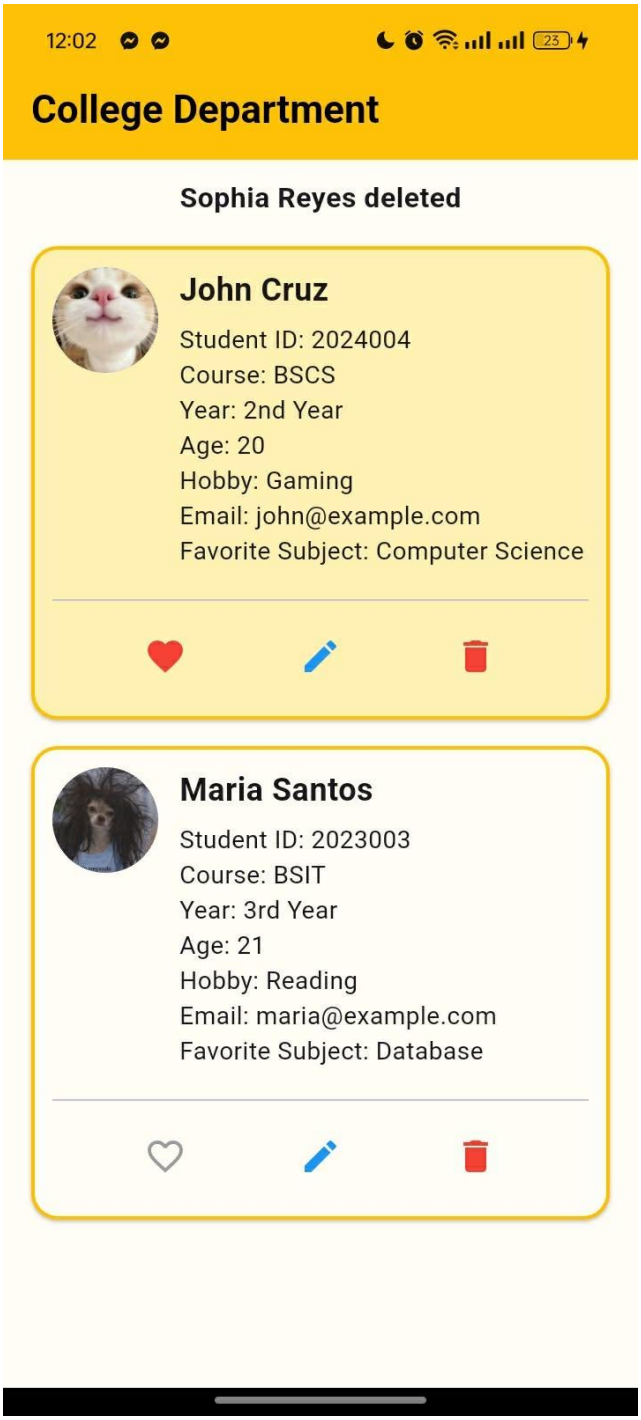
Your application should continue functioning correctly without affecting unrelated students.

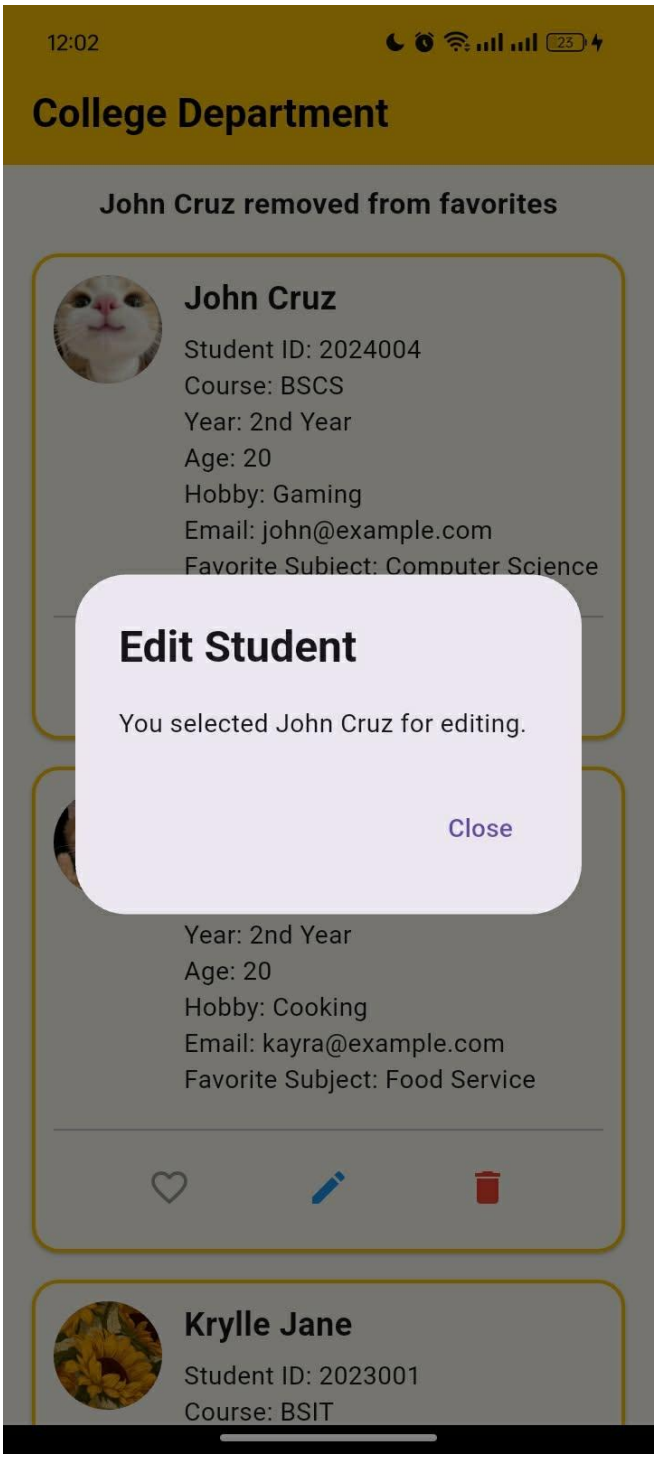
Test Checklist

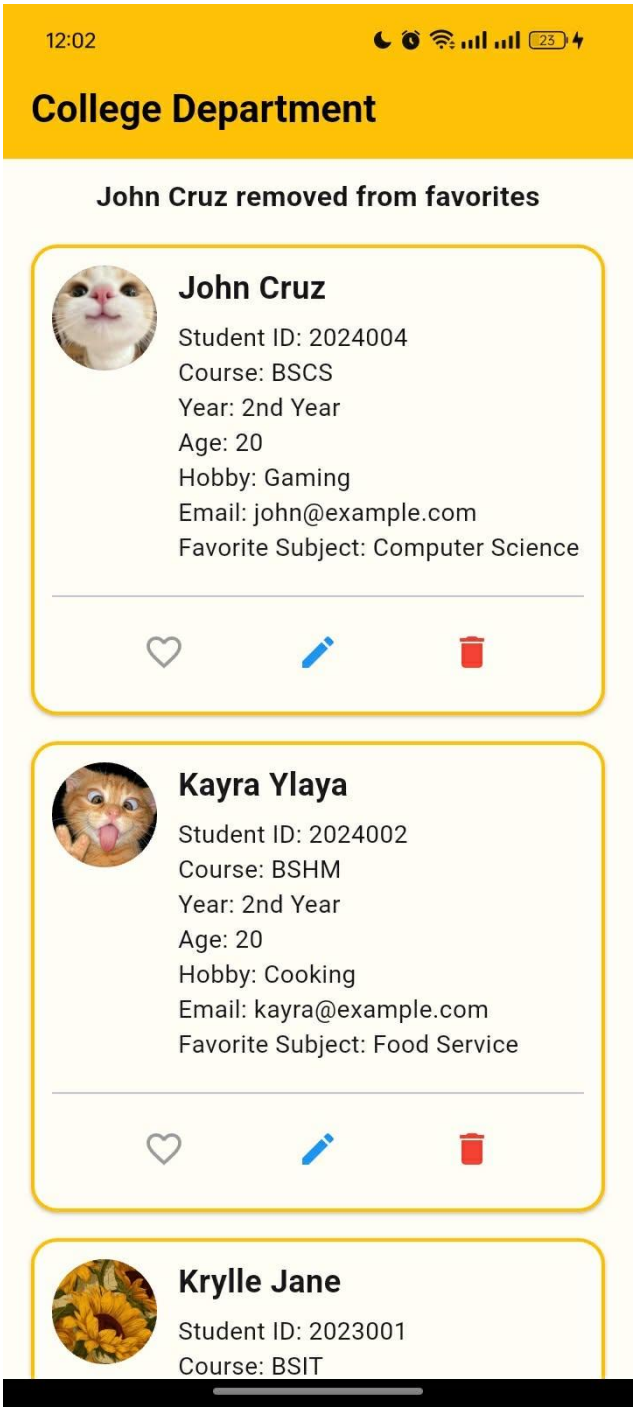
- Favorite works independently.
- Delete updates the list immediately.
- Edit still opens correctly.
- Remaining students keep their own states.

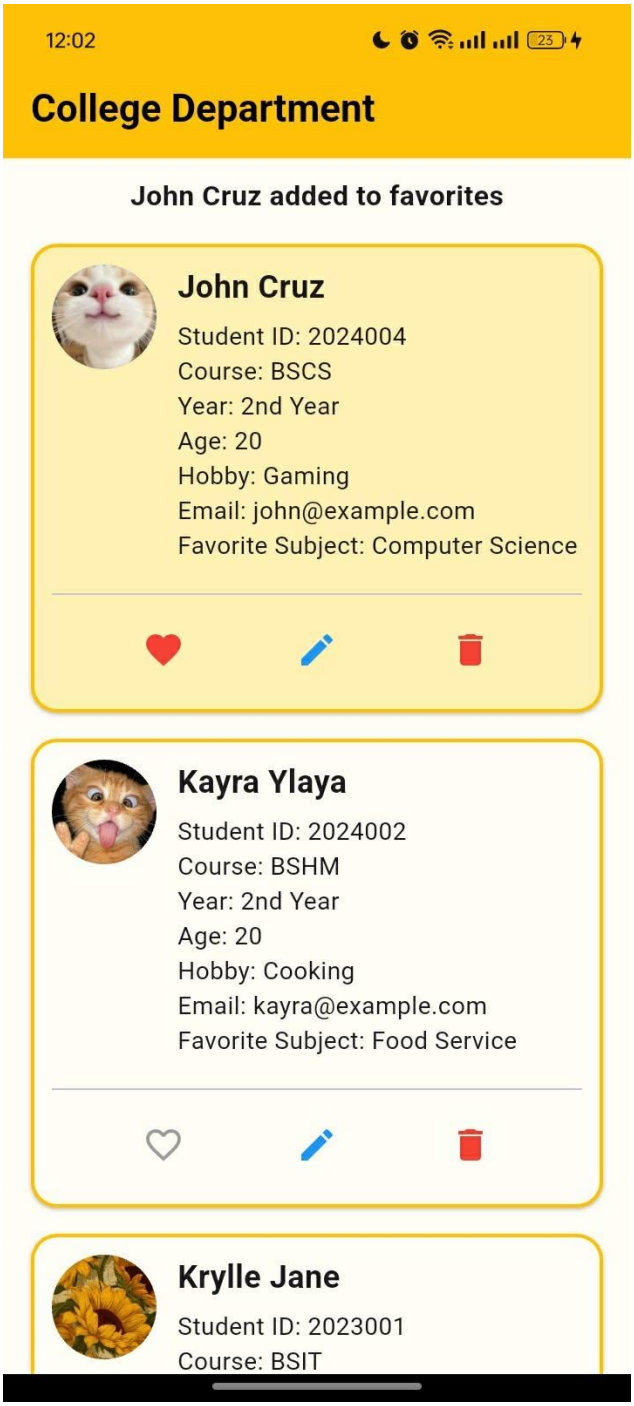
Screenshot:

- Multiple interactions working together
- Application still functioning normally


12:02
College Department
Sophia Reyes deleted
John Cruz
Student ID: 2024004
Course: BSCS
Year: 2nd Year
Age: 20
Hobby: Gaming
Email: john@example.com
Favorite Subject: Computer Science
Maria Santos
Student ID: 2023003
Course: BSIT
Year: 3rd Year
Age: 21
Hobby: Reading
Email: maria@example.com
Favorite Subject: Database


12:02
College Department
John Cruz removed from favorites
John Cruz
Student ID: 2024004
Course: BSCS
Year: 2nd Year
Age: 20
Hobby: Gaming
Email: john@example.com
Edit Student
You selected John Cruz for editing.
Close
Year: 2nd Year
Age: 20
Hobby: Cooking
Email: kayra@example.com
Favorite Subject: Food Service
Krylle Jane
Student ID: 2023001
Course: BSIT


12:02
College Department
John Cruz removed from favorites
John Cruz
Student ID: 2024004
Course: BSCS
Year: 2nd Year
Age: 20
Hobby: Gaming
Email: john@example.com
Favorite Subject: Computer Science
Kayra Ylaya
Student ID: 2024002
Course: BSHM
Year: 2nd Year
Age: 20
Hobby: Cooking
Email: kayra@example.com
Favorite Subject: Food Service
Krylle Jane
Student ID: 2023001
Course: BSIT


12:02
College Department
John Cruz added to favorites
John Cruz
Student ID: 2024004
Course: BSCS
Year: 2nd Year
Age: 20
Hobby: Gaming
Email: john@example.com
Favorite Subject: Computer Science
Kayra Ylaya
Student ID: 2024002
Course: BSHM
Year: 2nd Year
Age: 20
Hobby: Cooking
Email: kayra@example.com
Favorite Subject: Food Service
Krylle Jane
Student ID: 2023001
Course: BSIT

Flag 8 — Save Your Progress

Objective: Commit and push your completed Interactive Student Directory.

Use a proper Git commit message.

Branch:

`feature/student-interactions`

Screenshot:

- Successful commit
- Current branch
- Terminal showing the commit

